

**SafeCheck–SignSpeak: An Integrated RFID-Based Student
Attendance, Safety Monitoring, and Filipino Sign Language
Recognition System for Philippine School for the Deaf**

A capstone project proposal
Presented to the faculty of the
Information and communication Technology Program
STI College Alabang

In partial Fulfillment
Of the Requirements of the Degree
Bachelor of Science in Information Technology

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: SafeCheck–SignSpeak: An Integrated RFID-Based Student Attendance, Safety Monitoring, and Filipino Sign Language Recognition System for Philippine School for the Deaf

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for the degree Bachelor of Science in Information Technology
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ABSTRACT

Title of research: **SafeCheck–SignSpeak: An Integrated RFID-Based Student Attendance, Safety Monitoring, and Filipino Sign Language Recognition System for Philippine School for the Deaf**

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The study was motivated by two critical gaps identified in the target institution: the absence of a real-time mechanism to inform parents and guardians of their child's arrival at and departure from the school premises, and the lack of an accessible and interactive platform to support deaf students in developing their Filipino Sign Language competencies. Traditional attendance methods were found to be manual, time-consuming, and prone to human error, while existing sign language systems and RFID-based attendance solutions addressed only singular functions without integration into a unified platform.

SafeCheck–SignSpeak employs RFID cards and readers installed at designated school entrance and exit points to automatically log student Time In and Time Out records, which are stored in a centralized PostgreSQL database hosted via Supabase. Upon each successful

RFID scan, the system triggers real-time push notifications to the registered mobile devices of parents or guardians through an Android mobile application. Additionally, gestures through camera input, providing students with immediate recognition feedback and interactive learning activities including practice, game, and quiz modes.

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LIST OF NOTATIONS

Notation	Meaning
API	Application Programming Interface
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
FSL	Filipino Sign Language
IDE	Integrated Development Environment
JWT	JSON Web Token
LRN	Learner Reference Number
LSTM	Long Short-Term Memory
ORM	Object-Relational Mapping
PSD	Philippine School for the Deaf
RFID	Radio Frequency Identification
SMS	Short Message Service
SVM	Support Vector Machine
UHF	Ultra High Frequency
UI	User Interface
SSD	Solid-State Drive

INTRODUCTION

Project Context

One of the concerns in Philippine schools for the deaf is the limited real-time visibility of parents and guardians regarding their child's arrival at and departure from the school premises. When parents are not immediately informed whether their child has already entered or left the campus, monitoring student safety during school hours becomes more difficult. According to Bati-On, Kuizon, Catulpos, Bonghanoy, and Agoylo (2025), RFID technology can be used in automated attendance management to improve the accuracy and efficiency of monitoring student presence in educational institutions. In relation to SafeCheck-SignSpeak, the RFID feature is applied specifically to record students' gate Time In and Time Out at the Philippine School for the Deaf. Through this feature, the system can provide timely push app notifications to parents or guardians whenever a student enters or exits the school premises, supporting parent awareness, student monitoring, and organized gate attendance records.

Recent studies demonstrated that the RFID-based attendance systems can improve accuracy, speed up processing time, and support real-time notification for parents. These systems automate the capturing of time-in and time-out events and can be integrated with notification features to support better monitoring of student presence. Although there are many existing implementations like these, they only focus primarily on attendance logging and do not include student safety monitoring, such as real-time visibility of student entry and exit from the campus and real-time notification for parents and guardians.

The use of RFID technology in school monitoring systems is supported by related studies that emphasize its capability to record student entry and attendance data accurately and efficiently. For instance, Kumar and Prasad (2021) presented an RFID-based student attendance monitoring model that records student attendance when RFID tags are scanned upon entry. This shows that RFID technology can be used as a reliable tool for capturing student presence through automated scanning. In relation to SafeCheck-SignSpeak, the RFID feature is applied specifically to monitor students'

gate Time-In and Time-Out at the Philippine School for the Deaf. Instead of focusing on classroom attendance or roll-checking, the system aims to provide parents and guardians with timely visibility of their child's arrival and departure from the school premises through push app notifications. This connection strengthens the relevance of RFID technology in supporting student monitoring, parent awareness, and school safety communication.

While in the deaf students, they face additional challenges, especially in communication and learning support. Students need to improve their learning while enjoying it. Learning Filipino Sign Language (FSL) plays an essential role in helping deaf students to understand lessons and interact with the school environment. One of the main problems is the limited accessibility of communication support within schools.

Prior works on sign language recognition and translation have explored the use of computer vision, deep learning, and transfer learning to recognize and interpret sign languages, including Filipino Sign Language (FSL), for educational and communication support. These studies highlight the potential of machine learning-based systems to provide automatic sign recognition, real-time feedback, and interactive learning experiences for deaf learners.

In summary, SafeCheck-SignSpeak enhances student safety and supports FSL learning in a single platform that adapts to the Philippine School for the Deaf. By combining RFID- based gate Time In/Time Out logging with real-time push app notifications to parents and a machine learning-based FSL recognition module, the proposed system delivers timely parent visibility into student arrival and departure notifications and a machine learning- based FSL recognition module, the proposed SafeCheck–SignSpeak system seeks to provide a comprehensive solution that addresses administrative efficiency, student safety, and the educational needs of deaf students. This integration aims to support teachers, administrators, parents/guardians, and learners through accurate attendance tracking, timely information on student presence within school premises, and interactive tools that encourage regular FSL practice.

Statement of the Problem

General Problems

The Parents and guardians at the Philippine School for the Deaf do not receive timely updates about when their child enters or leaves the school premises, limiting their visibility into student safety during school hours. These are time-consuming, prone to human error, and difficult to track over time. Since there's no system for faster attendance and real-time tracking for parents/guardians. Parents and Guardians do not receive real-time updates regarding their children's presence inside the school, limiting their awareness of student safety during school hours. Deaf students also face challenges in communication and learning support, particularly in developing Filipino Sign Language (FSL) skills in an engaging and consistent way. Within the school environment, there is no existing interactive and technology-driven platform that allows students to regularly practice FSL gestures, receive immediate recognition feedback, and assess their own progress. FSL learning remains largely dependent on traditional classroom instruction, which limits the students' opportunities for independent practice and skill development outside of direct teacher supervision.

Specific Problems

- **How can the system automatically and accurately record students' gate Time In and Time Out using RFID technology**

The system will use RFID cards and readers installed at the school entrance and exit points to automatically capture each student's tag ID and timestamp. It will then store the data in a centralized database and determine whether the scan is a Time In or Time Out record.

- **How can the system provide real-time push app notifications to parents or guardians regarding students' entry and exit from the school premises?**

The system will automatically send a push app notification to the parent or guardian's mobile device whenever a student taps their RFID card at the gate, informing them whether their child has entered or exited the school premises.

- **How can the system manage and maintain student records, RFID card assignments, and user accounts in a centralized and organized manner?**

The system will include an administration module where authorized personnel can register students, assign RFID cards, and manage user accounts. It will also support RFID card status updates such as active, lost, replaced, or suspended to keep all records accurate.

- **How can the system allow teachers to view, monitor, and track student gate attendance records through a dedicated teacher dashboard?**

The system will provide teachers with a dedicated dashboard where they can view and filter gate Time In and Time Out records by student name, section, or date, giving them organized visibility over student attendance within their classes.

- **How can a machine learning-based Filipino Sign Language recognition module detect and interpret basic FSL gestures appropriate for students?**

The system will use MediaPipe for hand landmark detection and a trained machine learning model for gesture classification. It will process the student's hand gesture through camera input and provide real-time recognition feedback.

- **How can a user-friendly interface help administrators view gate Time In and Time Out records and generate reports efficiently and securely?**

The system will provide a role-based web dashboard where authorized users can access, search, filter, and generate reports from gate attendance records, ensuring data is organized and protected from unauthorized access.

- **How can an interactive FSL learning feature encourage students to practice regularly and improve their gesture accuracy, confidence, and communication skills?**

The system will offer structured learning modes — lesson, practice, game, and quiz — where students perform FSL gestures in front of the camera and receive immediate feedback, with their progress and scores recorded throughout.

- **How can the system track and display students' FSL learning progress, scores, and activity history to monitor skill development over time?**

The system will record each student's completed lessons, quiz scores, and gesture performance results, which can be viewed by both the student and their teacher to monitor improvement and guide further FSL practice.

- **How can the system ensure the security of student data, attendance records, and user accounts against unauthorized access?**

The system will implement JWT-based authentication, password hashing, and role-based access control, ensuring that each user can only access data and features within their assigned role in compliance with data privacy regulations..

- **How can the system be made accessible across both web and mobile platforms to support the different needs of administrators, teachers, parents, and students?**

The system will be deployed as a web dashboard for administrators and teachers, and an Android mobile application for parents and students, ensuring all users can access the system's features conveniently from their preferred device.

- **How effective is the integrated SafeCheck-SignSpeak system in terms of functionality, usability, and its support for gate attendance monitoring, parent notification, and FSL learning?**

The system will be evaluated through system testing and user evaluation, measuring the accuracy of RFID logs, notification delivery, FSL recognition performance, and overall usability among the target users.

Purpose and Description of the Project

This study aims to develop SafeCheck–SignSpeak, an integrated RFID-based student attendance, safety monitoring, and Filipino Sign Language (FSL) recognition system specifically tailored for the Philippine School for the Deaf (PSD). The system was conceived in response to two critical and co-existing gaps identified within the institution: the absence of a real-time parent notification mechanism for student gate monitoring, and the lack of an interactive, technology-driven platform for Filipino Sign Language learning and assessment among deaf students.

In terms of attendance and safety monitoring, the system addresses the inability of parents and guardians to receive timely information regarding their child's arrival at and departure from the school premises. SafeCheck–SignSpeak employs RFID cards assigned to each student and RFID readers installed at designated school entrance and exit points. When a student taps their RFID card, the system automatically captures the tag ID and timestamp, records it as a gate Time In or Time Out entry in a centralized database, and immediately triggers a push notification to the parent or guardian's registered mobile device. This

feature aims to improve parental awareness, strengthen student safety monitoring, and eliminate the inefficiencies associated with manual attendance tracking.

In terms of Filipino Sign Language education, the system integrates a machine learning–based FSL recognition module designed to support deaf students in developing their sign language competencies in an engaging and consistent manner. The module utilizes a camera to capture the student's hand gestures, which are then processed by a trained deep learning model capable of recognizing basic FSL gestures and providing real-time recognition feedback. The system further supports learning through structured modes including lesson mode, practice mode, game mode, and quiz mode, each designed to encourage regular FSL practice and gradual skill improvement. Student progress, including completed letters, performance scores, and activity history, is tracked and recorded within the platform.

SafeCheck–SignSpeak is accessible through both a web dashboard and an Android mobile application, supporting four distinct user roles: Administrator, Teacher, Student, and Parent. By integrating RFID-based gate attendance monitoring, real-time parental push notifications, and an AI-powered FSL learning platform into a single unified system, SafeCheck–SignSpeak seeks to provide a comprehensive solution that enhances administrative efficiency, strengthens student safety visibility, and promotes inclusive communication development for deaf learners at the Philippine School for the Deaf.

Objectives

General Objectives

The general objective of this study is to develop SafeCheck–SignSpeak, an integrated RFID-based student attendance, safety monitoring, and Filipino Sign Language (FSL) recognition system designed for the Philippine School for the Deaf. The system aims to provide an efficient and reliable way to monitor students' attendance and ensure their safety within the school premises. In addition, SafeCheck–SignSpeak integrates a machine

Learning-based FSL recognition feature to support all students in improving their sign language skills through interactive learning. This platform ensures that attendance tracking, safety monitoring, and sign language practice are accessible, organized, and supportive of both academic development and student well-being.

Specific Objectives

- **To develop an RFID-based gate attendance system that automatically records students' Time In and Time Out.**

This will be achieved by integrating RFID cards and readers at designated entrance and exit points with a centralized database that automatically stores each scan as a Time In or Time Out record.

- **To implement a real-time push app notification feature that informs parents or guardians when a student enters or exits the school premises.**

This will allow parents or guardians to receive timely mobile notifications whenever their child taps their RFID card at the gate, improving parent awareness and student safety monitoring.

- **To develop a centralized account and records management module that handles student registration, RFID card assignments, and user account management.**

This will be accomplished by building an administration module that allows authorized personnel to register students, assign RFID cards, and manage user accounts with support for card status updates such as active, lost, replaced, or suspended.

- **To provide teachers with a dedicated dashboard that allows them to view, monitor, and track student gate attendance records efficiently.**

This will be achieved by developing a teacher-specific dashboard where teachers can search and filter gate attendance records by student name, section, or date within their assigned classes.

- **To integrate a machine learning-based Filipino Sign Language recognition module that detects and interprets basic FSL gestures.**

This will be done using MediaPipe for hand landmark detection and a trained machine learning model for gesture classification, providing students with real-time recognition feedback through camera input.

- **To develop a user-friendly interface that allows administrators to view gate Time In and Time Out records and generate reports efficiently and securely.**

This will be achieved by building a role-based web dashboard with search, filter, and report generation features, designed to be simple, organized, and secure for all authorized users.

- **To create an interactive FSL quiz examination that encourages students to practice regularly and improve their gesture accuracy, confidence, and communication skills.**

This will provide students with structured FSL learning modes — lesson, practice, game, and quiz where they perform gestures in front of the camera and receive immediate feedback, with progress and scores tracked throughout.

- **To implement a student FSL progress tracking feature that records learning activity, performance scores, and gesture history to monitor skill development over time.**

This will be accomplished by building a progress tracking module that stores completed lessons, quiz scores, and gesture performance results, accessible to both students and teachers for monitoring individual FSL development.

- **To implement a secure authentication and role-based access control system that protects student data, attendance records, and user accounts from unauthorized access.**

This will be achieved using JWT-based authentication, password hashing, and defined role permissions for each user type, ensuring all data is protected and accessible only to authorized users.

- **To ensure cross-platform accessibility of the system through a web dashboard and an Android mobile application to support all user roles.**

The system will be deployed as a web dashboard for administrators and teachers, and an Android mobile application for parents and students, ensuring consistent and convenient access for all users.

- **To evaluate the system's functionality, usability, and effectiveness in supporting gate attendance monitoring, parent notification, and FSL learning.**

This will be done through system testing and user evaluation measuring RFID scanning accuracy, notification reliability, FSL recognition performance, and overall usability among administrators, teachers, parents, and students.

Significance of the Project

The SafeCheck-SignSpeak system benefits multiple stakeholders:

- **The School/Institution** – The institution will benefit from a reliable and automated student attendance monitoring system that reduces manual record-keeping, minimizes errors, and ensures accurate, real-time tracking of student presence through RFID technology, resulting in improved administrative efficiency and data integrity.
- **Teachers and Faculty** – The teachers and faculty will benefit from an automated attendance logging system that eliminates the time-consuming process of manual roll calls, allowing them to dedicate more instructional time to teaching and focus on student engagement rather than administrative tasks.
- **Students** – The students of the Philippine School for the Deaf will benefit from an automated and accurate attendance logging system that ensures their presence is properly recorded, while the integrated sign language recognition feature bridges communication barriers by translating sign language gestures into text or speech, promoting inclusivity and enabling more meaningful participation in academic settings.
- **Parents and Guardians** – The parents and guardians will benefit from improved confidence in the institution's monitoring capabilities, knowing that their children's attendance is tracked accurately and systematically, fostering greater trust in the school's administrative processes.
- **The Education Sector** – The education sector will benefit from this study as it demonstrates the viability of integrating RFID technology and machine learning-based

sign language recognition into a unified system that enhances both administrative efficiency and inclusive education practices.

- **Future Developers and Researchers** – Future developers and researchers will benefit from this study as it provides a foundational reference for designing inclusive, technology-driven school management systems, and serves as a basis for further advancements in assistive technology, computer vision, and automated attendance solutions within educational environments.

Scope and Limitations

Scope

The study covers the design, development, and evaluation of SafeCheck–SignSpeak, an integrated RFID-based student attendance, safety monitoring, and Filipino Sign Language recognition system for the Philippine School for the Deaf. The system is composed of two integrated subsystems: the SafeCheck module, which handles RFID-based gate attendance and safety monitoring, and the SignSpeak module, which provides Filipino Sign Language (FSL) interactive learning and gesture recognition. The system is accessible through a web dashboard and mobile application, supporting multiple users, including Administrator, Teacher, Student, and Parent.

WEB APPLICATION

Landing Page and Authentication

- **Landing Page**

Displays system information and provides quick access to login options for all user roles.

- **Login Module**

Allows all users (Admin, Teacher, Student, Parent) to securely access the system using their credentials via JWT-based authentication.

Admin Portal

- **Dashboard**

Displays a real-time summary of system activity including total users, total students enrolled, recent RFID attendance logs, system status (RFID Scanner, AI Service, Database), and a user breakdown (students vs. staff).

- **Manage User**

Allows the admin to view, search, add, edit, and delete user accounts across all roles (Admin, Teacher, Student, Parent). Displays each user's name, username, email, role badge, grade level, RFID card status, and account creation date.

- **RFID Management**

Displays all students with their assigned RFID card numbers and statuses (Active or Unassigned). The admin can monitor RFID assignment and view recent scan activity logs statuses Time-in and Time-out.

- **FSL Module Progress**

Provides a system-wide view of student engagement with the FSL learning module. Displays per-student FSL accuracy rates, current words/letters being practiced, attempt counts, last practiced timestamps, and overall status (Completed, Learning, Needs Practice). Includes aggregate stats such as average AI accuracy, active learners, and completed modules.

- **Reports**

Displays attendance trend charts (using Recharts) showing daily attendance and FSL activity over the week. Also contains RFID activity logs with student names, actions, timestamps, and statuses.

- **System Configuration**

Allows the admin to manage academic settings (School Year, Semester), toggle hardware services (RFID Scanner, AI Vision), configure notification settings (SMS Alerts to Parents), and perform danger-zone actions (Reset Attendance Data).

Teacher Portal

- **Assessment (Phase 2 – Under Development)**

Planned feature for creating FSL quizzes and tracking student scores. Currently shows an under-development placeholder with a Phase 2 label.

- **My Profile**

Allows the teacher to update their first and last name (Personal Info tab) and change their account password (Security tab) with a strength indicator. Displays account details (Teacher ID, Username, Role).

Student Portal

- **Dashboard**

Displays a personalized greeting, today's attendance status (via RFID tap), stat cards (Attendance Rate, FSL Progress, Learning Streak, Quiz Best Score), a recent attendance list, FSL progress bar with letter count, and a quick-access section.

- **My Attendance**

Displays personal attendance statistics Time-in Time-Out and a full paginated attendance record table with date, time in, time out, and status. Includes month navigation and an export option.

- **FSL Learning**

Interactive Filipino Sign Language learning module. Students can practice FSL alphabet letters (A–Y, excluding J and Z which are motion-based). Each letter card shows the hand sign image/camera feed, and the AI model evaluates the

student's sign in real time for accuracy. Tracks progress, streaks, and completion per letter.

- **FSL Quiz**

A timed 10-question multiple-choice quiz on the FSL alphabet. Questions are randomized each session with a 20-second timer per question. Displays score, answer review with explanations, and tracks the student's personal best score.

- **FSL Game**

A gamified Filipino Sign Language activity module designed to reinforce letter recognition through speed-based competition. Students race against each other in real time, where the winner is determined by who performs the correct FSL hand sign the fastest.

- **My Profile**

Allows students to update their name and change their password. Displays account details (Student ID, Username, Grade Level, RFID Card status).

Parent Portal

- **Dashboard**

Displays a greeting, today's status today's status arrived at school/ left school for the linked child with check-in time, stat cards (Attendance Rate, FSL Progress, Learning Streak, Achievements), a recent attendance list, a mini FSL progress bar, and recent announcements from the teacher.

- **Attendance History**

Displays the linked child's full attendance record with statistics Time-In and Time-out a calendar view placeholder, and a records table showing date, time in, time out, and status. Includes month navigation and export.

- **FSL Progress**

Displays the child's FSL learning progress including letters mastered, overall percentage, current streak, and achievements/badges. Shows a weekly activity bar chart, an achievements panel (First Sign, Half the Alphabet, Week Streak, Full Alphabet), and an FSL Alphabet Map grid showing which letters are completed vs. pending. Letters J and Z are marked as locked/motion-based.

- **Report**

Allows the parent to select a month and year to generate a monthly performance report for their child. Displays attendance stats Time-in and Time-out, FSL stats (progress %, letters completed), and AI-generated notes (positive observations, warnings, informational insights). Includes a Download PDF option.

- **Notifications**

Displays all notifications for the parent filtered by type (Announcement, Attendance, FSL, System). Unread notifications are highlighted. Supports Mark All as Read. Each notification shows the type icon, title, message, sender, timestamp, and read status.

- **My Profile**

Allows the parent to update their name and change their password. Displays account details (Parent ID, Username, Role badge). Uses the same tabbed Personal Info / Security layout as other portals.

Supporting Components

- **RFID Attendance System**

Physical RFID scanners at school entry points communicate with the NestJS backend API. When a student taps their assigned RFID card, the system logs the time in and auto-determines the attendance status: Present (before or at 8:00 AM), Late (after 8:00 AM), or Absent (no tap by end of day). The Arduino-based hardware interfaces with the NestJS API through a network connection.

- **AI Powered FSL Recognition (SignSpeak Module)**

A machine learning model trained on Filipino Sign Language alphabet images (A–Y, excluding J and Z). The model is integrated into the Student portal's FSL Learning page via a camera feed. When a student performs a hand sign, the model evaluates and scores the accuracy in real time. The model was trained using Python with data augmentation to improve accuracy across different skin tones, lighting, and hand angles.

- **Authentication & Role-Based Access Control**

All portals use JWT-based authentication. Role guards on both the frontend (Next.js route protection) and backend (NestJS guards) ensure that each user can only access their designated portal. Unauthorized access redirects to the login page.

MOBILE APPLICATION

SCOPE

Student Portal (Mobile Application)

- **Dashboard**

Displays a personalized greeting, today's attendance status (via RFID tap), stat cards (Attendance Rate, FSL Progress, Learning Streak, Quiz Best Score), a recent attendance list, FSL progress bar with letter count, and a quick-access section.

- **My Attendance**

Displays personal attendance statistics Time-in Time-Out and a full paginated attendance record table with date, time in, time out, and status. Includes month navigation and an export option.

- **FSL Learning**

Interactive Filipino Sign Language learning module. Students can practice FSL alphabet letters (A–Y, excluding J and Z which are motion-based). Each letter card shows the hand sign image/camera feed, and the AI model evaluates the student's sign in real time for accuracy. Tracks progress, streaks, and completion per letter.

- **FSL Quiz**

A timed 10-question multiple-choice quiz on the FSL alphabet. Questions are randomized each session with a 20-second timer per question. Displays score, answer review with explanations, and tracks the student's personal best score.

- **FSL Game**

A gamified Filipino Sign Language activity module designed to reinforce letter recognition through speed-based competition. Students race against each other in real time, where the winner is determined by who performs the correct FSL hand sign the fastest.

- **My Profile**

Allows students to update their name and change their password. Displays account details (Student ID, Username, Grade Level, RFID Card status).

Parent Portal (Mobile Application)

- **Dashboard**

Displays a greeting, today's status today's status arrived at school/ left school for the linked child with check-in time, stat cards (Attendance Rate, FSL Progress, Learning Streak, Achievements), a recent attendance list, a mini FSL progress bar, and recent announcements from the teacher.

- **Attendance History**

Displays the linked child's full attendance record with statistics Time-In and Time-out a calendar view placeholder, and a records table showing date, time in, time out, and status. Includes month navigation and export.

- **FSL Progress**

Displays the child's FSL learning progress including letters mastered, overall percentage, current streak, and achievements/badges. Shows a weekly activity bar chart, an achievements panel (First Sign, Half the Alphabet, Week Streak, Full Alphabet), and an FSL Alphabet Map grid showing which letters are completed vs. pending. Letters J and Z are marked as locked/motion-based.

- **Report**

Allows the parent to select a month and year to generate a monthly performance report for their child. Displays attendance stats Time-in and Time-out, FSL stats (progress %, letters completed), and AI-generated notes (positive observations, warnings, informational insights). Includes a Download PDF option.

- **Notifications**

Displays all notifications for the parent filtered by type (Announcement, Attendance, FSL, System). Unread notifications are highlighted. Supports Mark All as Read. Each notification shows the type icon, title, message, sender, timestamp, and read status.

- **My Profile**

Allows the parent to update their name and change their password. Displays account details (Parent ID, Username, Role badge). Uses the same tabbed Personal Info / Security layout as other portals.

Limitations of the Project

- The system is limited to RFID-based attendance monitoring within designated scanning areas inside the school premises only.
- The system requires an internet connection
- The system's performance depends on the proper functioning of RFID hardware; lost, damaged, or forgotten cards may affect attendance accuracy.
- The system was tested within a controlled school environment and is not validated for large-scale implementation across multiple institutions.
- The study focuses specifically on students of the Philippine School for the Deaf
- The RFID-based gate attendance component of the system is limited to recording students' arrival and departure times within the school premises, specifically at locations where the RFID scanner is installed.
- The FSL recognition module is limited to recognizing the 24 static alphabet letters of the Filipino Sign Language (FSL), specifically letters A to Y, excluding the letters J and Z, which require dynamic hand movements and are not supported in the current version of the system.
- Only Push App Notifications via the Android mobile app are implemented.
- The accuracy of the FSL recognition module may vary depending on environmental conditions such as lighting quality, camera resolution, background noise, and the hand positioning of the user during gesture performance.
- The mobile application is currently optimized for Android devices, ensuring stable performance and compatibility across a wide range of Android smartphones. Support for iOS devices is not included in the current version, but may be considered in future enhancements.

REVIEW OF RELATED LITERATURE

This chapter presents the related literature and studies that serve as the foundation of the proposed Safe-Check system. The reviewed materials focus on two main components: RFID-based student attendance and safety monitoring systems, and Filipino Sign Language (FSL) recognition using machine learning. Both local and foreign studies were examined to identify existing technologies, methodologies, and limitations, which help establish the relevance of the proposed system and justify the need for an integrated, cross-platform solution designed for the Philippine School for the Deaf.

Review of Related Literature (Local)

Lecturer and Student Attendance System with RFID (2021)

Renaldo et al. (2021) developed a real-time RFID-based attendance recording system designed for use in a university setting in the Philippines. The study addresses the limitations of traditional attendance systems, which are often manual, time-consuming, and prone to errors such as incorrect recording or proxy attendance. The researchers proposed an automated system that uses Radio Frequency Identification (RFID) technology together with a microcontroller and database system to improve attendance monitoring efficiency in higher education institutions. In the developed system, both lecturers and students are issued RFID-enabled identification cards. When the card is scanned using an RFID reader connected to the system, the unique ID of the user is automatically captured and transmitted to a computer system. The system then records the attendance in real time and stores the data in a centralized database. This allows administrators and instructors to easily monitor attendance records, generate reports, and track attendance history without the need for manual encoding.

Automated Attendance Management with RFID and Geospatial Visualization (2025)

According to Bati-On, Kuizon, Catulpos, Bonghanoy, and Agoylo (2025), an automated attendance management system was developed using Radio Frequency Identification (RFID) technology combined with geospatial visualization to enhance the accuracy and efficiency of attendance monitoring in educational institutions. The system utilizes RFID cards to automatically record student attendance, minimizing errors associated with manual checking and reducing administrative workload. The inclusion of geospatial visualization allows administrators to monitor student presence within specific locations in real time, improving oversight and data analysis. The study concluded that the system significantly improved attendance reliability, streamlined record management, and provided valuable insights for school administrators, supporting the effectiveness of RFID- based attendance solution.

Filipino Sign Language Translation through Transfer Learning (2024)

Tipan et al. (2024) proposed a Filipino Sign Language translation system utilizing transfer learning to improve gesture recognition accuracy while minimizing the need for large training datasets. By fine-tuning pre-trained deep learning models on Filipino Sign Language data, the system was able to translate hand gestures into corresponding textual outputs efficiently. The study showed that transfer learning significantly enhanced model performance and reduced computational requirements. This research emphasizes the practicality of applying advanced AI techniques to develop scalable and resource-efficient Filipino Sign Language recognition systems.

KUMPAS: A Filipino Sign Language Translator (2023)

Bautista, Cruz, Esbieto, Montemayor, and Panganiban (2023) of First City Providential College developed KUMPAS, a bilingual FSL translator system featuring two communication modes: sign-to-text and speech-to-sign. The system utilized MediaPipe for hand tracking and gesture detection, integrating face recognition and object detection features to improve recognition accuracy. Evaluation using ISO 25010 standards yielded an overall "Very Satisfactory" rating, confirming its practical usability and reliability as an FSL communication bridge for Filipinos.

Filipino Sign Language Recognition using Deep Learning (2021)

According to Montefalcon, Padilla, and Llabanes Rodriguez (2021), a Filipino Sign Language (FSL) recognition system was developed using deep learning techniques to address communication challenges faced by Deaf individuals in the Philippines. The study employed convolutional neural networks (CNN) to classify hand gestures captured through image-based input. The system achieved a high level of accuracy in recognizing common Filipino Sign Language gestures, demonstrating the effectiveness of deep learning in visual gesture interpretation. The results highlight the potential of artificial intelligence in supporting assistive technologies for Deaf communication and learning.

Review of Related Literature (Foreign)

I-Attend: A Smart Attendance System for Secondary Schools Using RFID (2025)

Singh et al. (2025) introduced I-Attend, a smart attendance system designed for secondary schools using RFID technology. The system enables automatic identification and attendance recording of students through RFID cards, ensuring fast and accurate

data collection. The study also emphasized system scalability and ease of integration with existing school infrastructure. Results showed that the RFID-based system enhanced attendance management efficiency and improved monitoring capabilities. This research supports the implementation of RFID technology as a reliable tool for attendance tracking and student monitoring in school environments.

Sign Language Interpretation Using Machine Learning and Artificial Intelligence (2024)

According to Najib (2024), a sign language interpretation system was developed using machine learning and artificial intelligence techniques to convert hand gestures into understandable text and speech outputs. The study focused on improving recognition accuracy through advanced classification algorithms and feature extraction methods. Results indicated that AI-based models significantly enhanced gesture recognition performance, enabling more efficient communication for Deaf and mute individuals. The study highlights the growing role of artificial intelligence in assistive communication technologies and supports the feasibility of implementing intelligent sign language interpretation systems in real-world applications.

A Foreign Related Study: Gesture Recognition of Filipino Sign Language Using CNN-LSTM Deep Neural Networks (2024)

This The study Gesture Recognition of Filipino Sign Language Using Convolutional and Long Short-Term Memory Deep Neural Networks by Karl Jensen Cayme et al. (2024) introduced a machine-learning-based system capable of recognizing gestures used in Filipino Sign Language. (FSL). The researchers developed a recognition model that combines Convolutional Neural Networks (CNN) and Long Short-Term Memory

(LSTM) networks to analyze visual features and sequential hand movements captured through video input. The system processes images through several preprocessing stages such as segmentation, normalization, and feature extraction before passing them to the deep learning model for gesture classification. The developed model was able to recognize several FSL expressions and transactional gestures in real time, achieving high accuracy in identifying hand movements and patterns. The researchers emphasized that the integration of computer vision and machine learning technologies can significantly improve communication between deaf individuals and the hearing community.

Real-Time Sign Language Detection (2022)

Yadav et al. (2022) developed a real-time sign language detection system using computer vision and machine learning techniques to recognize hand gestures instantly. The system processes live video input and applies image segmentation and gesture classification to achieve fast and accurate detection. Experimental results showed that the system performed efficiently under real-time conditions, making it suitable for practical deployment. This study supports the integration of real-time sign language recognition technologies, which aligns with the objectives of implementing responsive and interactive sign language systems in assistive and educational platforms.

A New Model of the Student Attendance Monitoring System Using RFID Technology (2021)

According to Kumar and Prasad (2021), a new model for a student attendance monitoring system was developed using Radio Frequency Identification (RFID)

technology to improve accuracy and efficiency in educational institutions. The system automatically records student attendance when RFID tags are scanned upon entry, eliminating the need for manual attendance checking. The study reported that the RFID-based system reduced attendance errors, saved instructional time, and improved record reliability. The researchers concluded that RFID technology is an effective solution for managing large scale student attendance systems.

SYNTHESIS

The reviewed local and foreign studies demonstrate the increasing use of advanced technologies in enhancing educational accessibility, communication, and monitoring systems. Local studies on Filipino Sign Language (FSL) recognition emphasize the effectiveness of deep learning, convolutional neural networks, and transfer learning techniques in accurately recognizing hand gestures and translating them into text outputs. These findings confirm the feasibility of implementing AI-based assistive technologies tailored to the Filipino Deaf community. Similarly, foreign studies strengthen these findings by presenting scalable and real-time sign language recognition systems that improve communication through machine learning and artificial intelligence applications. In terms of attendance and safety monitoring, both local and foreign studies consistently show that Radio Frequency Identification (RFID) technology improves the accuracy and efficiency of student attendance system.

Local implementations highlight the integration of RFID with notification systems and data analytics to enhance monitoring capabilities, while foreign systems validate the reliability and scalability of RFID in managing large student populations. These studies confirm that RFID technology is a dependable solution for automating attendance tracking and strengthening institutional monitoring processes. Despite the effectiveness of sign language recognition systems and RFID-based attendance systems when implemented independently, most existing solutions

focus on only one function. There is limited integration of both communication accessibility and safety monitoring within a single platform, particularly in Deaf educational institutions in the Philippines. Therefore, the proposed SafeCheck–

SignSpeak system addresses this gap by integrating RFID-based attendance and safety monitoring with Filipino Sign Language recognition into one unified system. By combining these technologies, the study aims to enhance institutional efficiency while promoting inclusive communication and improved student safety.

TECHNICAL BACKGROUNDS

Overview of Current Technologies to be used in the System

The SafeCheck-SignSpeak system is developed as a cross-platform web and mobile-based application designed for attendance monitoring and Filipino Sign Language (FSL) learning. The system integrates modern web, and mobile technologies to ensure efficient performance, real-time processing, and a user-friendly experience. It utilizes a combination of frontend frameworks, backend services, cloud databases, and machine learning tools to deliver its core functionalities.

TypeScript & JavaScript

The developers use TypeScript and JavaScript as the primary programming languages for both frontend and backend development. These technologies enable structured, scalable, and maintainable code, while supporting dynamic and interactive features across the system.

HTML5 & CSS (Tailwind CSS)

HTML5 is used to structure the content of the web application, ensuring proper layout and accessibility across modern browsers. CSS, particularly Tailwind CSS, is used to design and style the user interface, allowing responsive layouts, consistent design, and improved user experience.

Next.js 16

The developers use Next.js as the main web frontend framework to build the system's dashboards for Admin, Teacher, Student, and Parent. It enables fast rendering, efficient routing, and optimized performance for modern web applications.

React & React Native (Expo)

React is used for building interactive user interfaces on the web, while React Native (with Expo) is used to develop the mobile application. This enables a consistent user experience between the web dashboard and the Android mobile application.

NestJS (Node.js)

NestJS is used as the backend framework to handle business logic, API development, authentication, and data processing. It ensures a scalable and well-structured server-side architecture.

PostgreSQL (Supabase) & Prisma ORM

PostgreSQL, hosted via Supabase, is used as the system's primary database for storing user data, attendance records, and learning progress. Prisma ORM is used to manage database interactions efficiently, ensuring secure and structured data handling.

Python FastAPI

Python FastAPI is used to develop the machine learning server responsible for processing FSL recognition. It handles requests from the main system and returns prediction results in real time.

MediaPipe & Scikit-learn (SVM)

MediaPipe is used for hand landmark detection in real-time video input, while Scikit-learn's Support Vector Machine (SVM) is used to classify static FSL alphabet gestures. These technologies enable accurate and efficient letter recognition.

TensorFlow / Keras (LSTM)

TensorFlow and Keras are used to implement Long Short-Term Memory (LSTM) models for dynamic FSL word recognition. This allows the system to analyze gesture sequences and recognize complete words.

JWT & Passport.js

The system uses JSON Web Tokens (JWT) and Passport.js for authentication and security. These technologies ensure secure login, session management, and role-based access control across different users.

Expo Notifications

Expo Notifications API is used to send real-time push app notifications to parents whenever a student taps their RFID card at the school gate (Time In/Time Out). This ensures immediate communication between school and home.

Git & GitHub

Git and GitHub are used for version control and collaboration. These tools allow the developers to track changes, manage source code, and work efficiently as a team throughout the development process.

Vercel & Railway (Cloud Deployment)

The system is deployed using Vercel for the web frontend and Railway for the backend services. These platforms provide reliable cloud hosting, scalability, and continuous deployment capabilities.

OpenCV & NumPy

OpenCV and NumPy are used in the machine learning pipeline for image processing and numerical computations, supporting accurate gesture recognition and data processing.

Calendar of Activities

Brainstorming

During this phase, the proponents conducted collaborative discussions to identify the problems encountered at the Philippine School for the Deaf regarding student attendance monitoring and Filipino Sign Language learning support. The proponents discussed possible solutions, planned the proposed system features, and identified the scope and objectives of the study. This phase was conducted from the first week of February until the third week of April.

Interview

During the third week of February, the proponents conducted interviews and consultations with the school administration and staff of the Philippine School for the Deaf for information gathering regarding the current process gate management and Filipino Sign Language learning. The gathered information was used as a basis for identifying the problems and requirements of the proposed system.

Documentation: Chapter I

The proponents prepared Chapter I from the second week of February until the third week of March. This chapter includes the Project Context, Interview, Statement of the Problem, Objectives of the Study, and Scope and Limitations. Minor and major revisions were also applied based on the recommendations of the adviser.

Documentation: Chapter II

The proponents prepared Chapter II from the third week of March until the second week of April. This chapter includes the Review of Related Literature, Review of Related

Studies and Systems, and Synthesis. Related local and foreign studies were gathered and analyzed to support the relevance and feasibility of the proposed system.

Documentation: Chapter III

The proponents prepared Chapter III from the third week of April until the first week of May. This chapter includes the Overview of Current Technologies to be Used in the System, Calendar of Activities, and other supporting sections related to the technical background and methodology of the study.

Entity Relationship Diagram

The proponents created the Entity Relationship Diagram (ERD) from the third week of March until the third week of April. This includes revisions and improvements made to properly present the database structure, table relationships, and data flow of the SafeCheck-SignSpeak system.

Data Flow Diagram

The proponents created the Data Flow Diagram (DFD) from the third week of April until the first week of May. This diagram illustrates how data moves throughout the SafeCheck-SignSpeak system, showing the flow of information between the system's processes, data stores, and external entities such as the Administrator, Teacher, Student, and Parent, ensuring a clear representation of the system's data processing and communication structure.

Flowcharts

The proponents created the system flowcharts from the third week of March until the first week of April. The flowcharts illustrate the step-by-step process of the system's core functions, including RFID-based gate Time In and Time Out recording, push app

notification delivery to parents or guardians, and the Filipino Sign Language recognition and interactive learning flow for each user role.

Gantt Chart of Activities

Month	February				March				April				May				June			
Activity																				
Brainstorming																				
Documentation																				
Calendar of Activities																				
Project Context																				
Interview																				
Statement of the Problem																				
Objectives of the Study																				
Scope and Limitation																				
Review Related Literature/System																				
Synthesis																				
Overview of Current Technologies to be used in the system																				
Methodology, Results, and Discussion																				
Design of Software, System, Product, and/or Processes																				

Flowchart																				
Entity Relationship Diagram																				
Data Flow Diagram																				

Table 1. Gannt C

RESOURCES

Software Requirements	
Operating System	Windows 10 or higher
Mobile Development	React Native (Expo SDK 6)
Backend	NestJS (Node.js), REST API
Database	PostgreSQL (Supabase)
Programming Languages	TypeScript, JavaScript, Python, SQL
Browser	Google Chrome, Microsoft Edge, Firefox
Documentation	Microsoft Word, Google Docs

Table 2. Software Requirements for Development

Hardware Requirements	
Processor	Intel Core i5 / Ryzen 5 or higher
RAM	8GB or higher
Storage	512GB SSD or higher
Display	1366x768 or higher
RFID Device	720p Webcam (for FSL recognition)
Camera	Cam 1080p 30fps or above

Table 3. Hardware Resources for Development

Minimum Hardware Requirements (For Development)	
Processor	Intel Core i3 (8th Gen)
Memory	4GB or higher
Storage	256GB SSD
Camera	Basic webcam

Table 3.1 Minimum Hardware requirements for development

Minimum Hardware Requirements (For Users)	
Processor	Intel Core i3 (8th Gen)
Memory	4GB or higher
Storage	256GB SSD
Device	Android Smartphone (Android 10 or higher)

Table 3.2 Minimum Hardware requirements for users

METHODOLOGY, RESULTS, AND DISCUSSIONS

In For the development of this system, the Waterfall methodology was selected for developing SafeCheck-SignSpeak due to its linear and sequential structure, which ensures each phase is completed before proceeding to ideal project like RFID attendance, safety monitoring, and FSL recognition at the Philippine School for the Deaf. This approach minimizes risks in hardware- software integration, like RFID readers with RFID cards, it provides clear milestones for stakeholder approval, and produces comprehensive documentation essential for educational institution deployment, where changes post-requirements are costly and rare. Unlike agile methods, Waterfall's predictability suits the project's stable scope, regulatory needs for data privacy in student records, and one-time development for a specific school environment

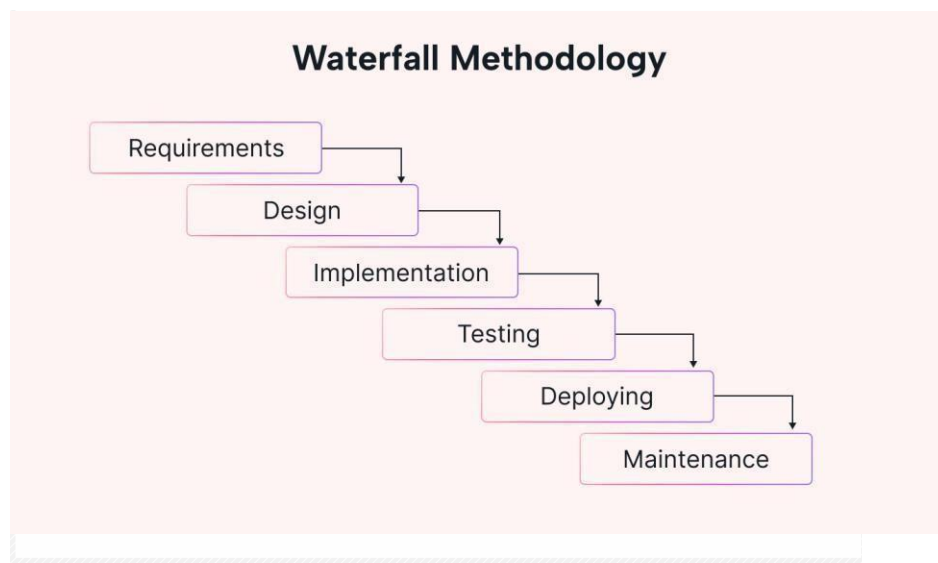


Figure 1. Waterfall Methodology Diagram

Stage 1: Requirements Gathering

We gathered the needed information and identified the requirements of the system. This includes understanding the problems in attendance Time-in Time-out, student safety, and FSL learning. We also reviewed related studies and consulted with our adviser to make sure the system meets the project objectives.

Stage 2: Design

Based on the gathered requirements, we designed the system architecture, database structure, and user interface.

Stage 3: Implementation

In this phase, we developed the system by coding the different modules. This includes the attendance gate Time-in Time-out system, notification feature, user management, and FSL recognition module. We also created and connected the database to store and manage system data. All system components were integrated to form a complete and working system.

Requirements Analysis

In this phase, the researchers analyzed the specific needs and functionalities required for the SafeCheck-SignSpeak system. This involved identifying the essential requirements needed to develop an RFID-based gate Time In and Time Out monitoring system with push app notifications for parents or guardians, together with a Filipino Sign Language quiz feature for students. The researchers carefully examined the requirements gathered from the client interview to ensure that the system would focus on monitoring students' entry and exit from the school premises. The team also considered the needs of different users, including administrators, teachers, parents or guardians, and students. This phase helped the researchers define the main features, system limitations, user roles, hardware requirements, and software requirements needed for the development of the system.

Requirements Documentation

In this phase, the team documented the identified functional and non-functional requirements necessary for the development of the SafeCheck-SignSpeak system.

This step ensured that the system objectives, features, and limitations were clearly defined and aligned with the updated client requirements. The documentation included the system modules, such as RFID gate Time In and Time Out tracking, push app notifications, user account management, student records, RFID card status management, FSL gesture recognition, interactive FSL learning modes, and FSL progress tracking. It also outlined the roles and responsibilities of each user, including administrators, teachers, parents or guardians, and students. This document served as a reference for the design, development, testing, and implementation phases of the system.

Design of Software, System, Product, and/or Processes

In this phase, the team documented the identified functional and non-functional requirements necessary for the development of the SafeCheck-SignSpeak system. This step ensured that the system objectives, features, and limitations were clearly defined and aligned with the updated client requirements. The documentation included the system modules, such as RFID gate Time In and Time Out tracking, push app notifications, user account management, student records, RFID card status management, FSL gesture recognition, interactive FSL learning modes, and FSL progress tracking. It also outlined the roles and responsibilities of each user, including administrators, teachers, parents or guardians, and students. This document served as a reference for the design, development, testing, and implementation phases of the system.

Flow Chart

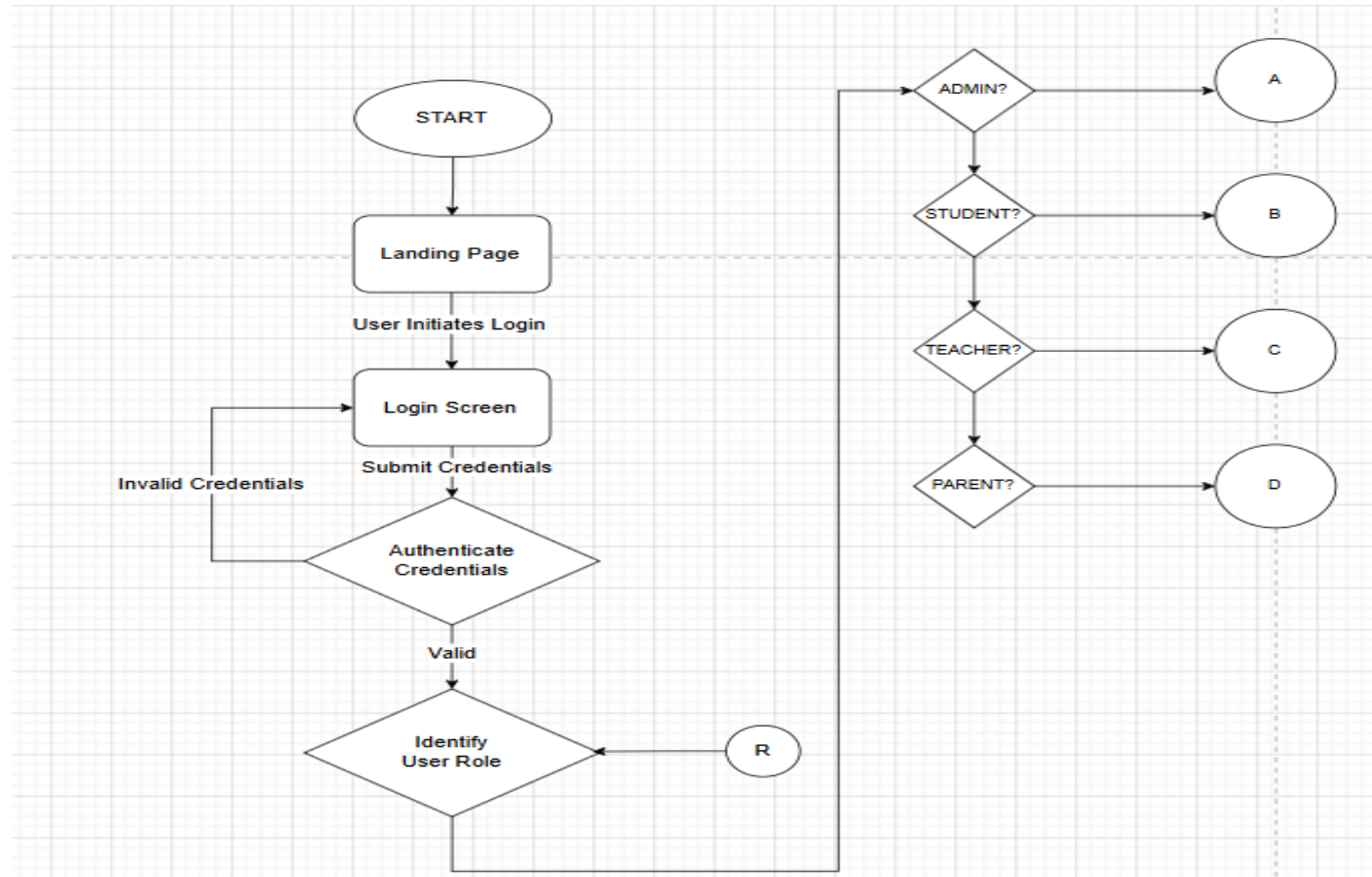


Figure 2 Flowchart (Login)

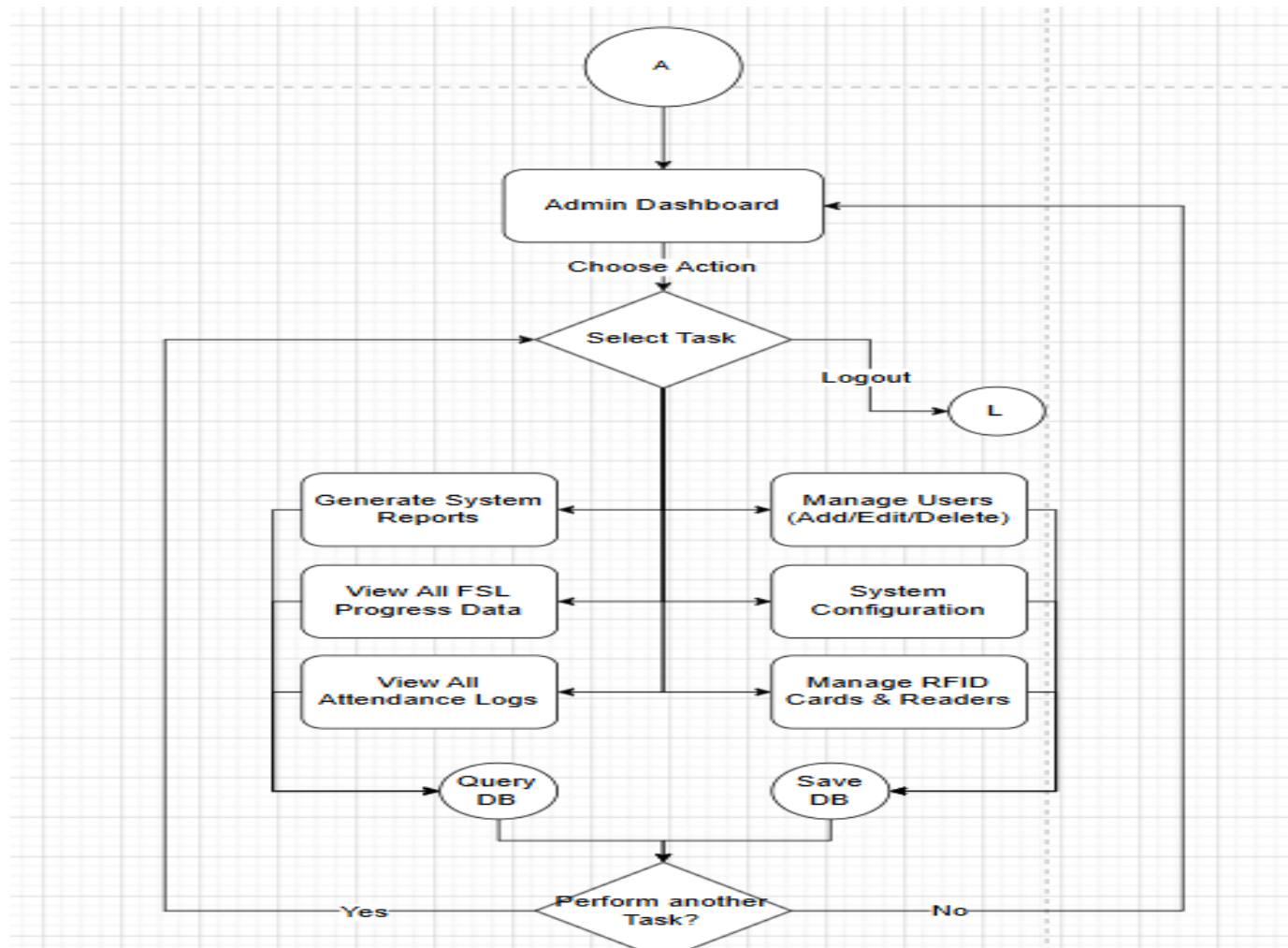


Figure 3 Flowchart (Admin)

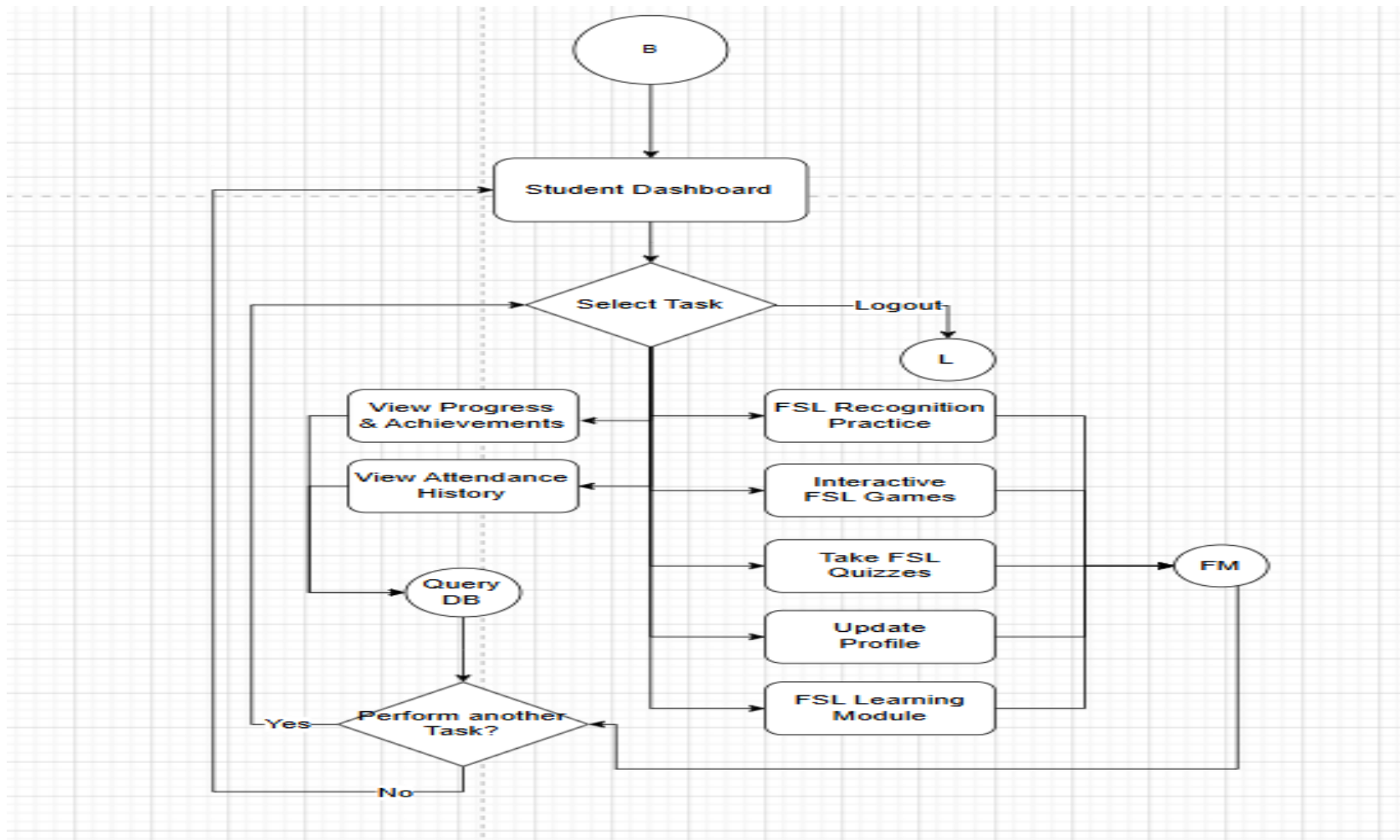


Figure 4 Flowchart (Student)

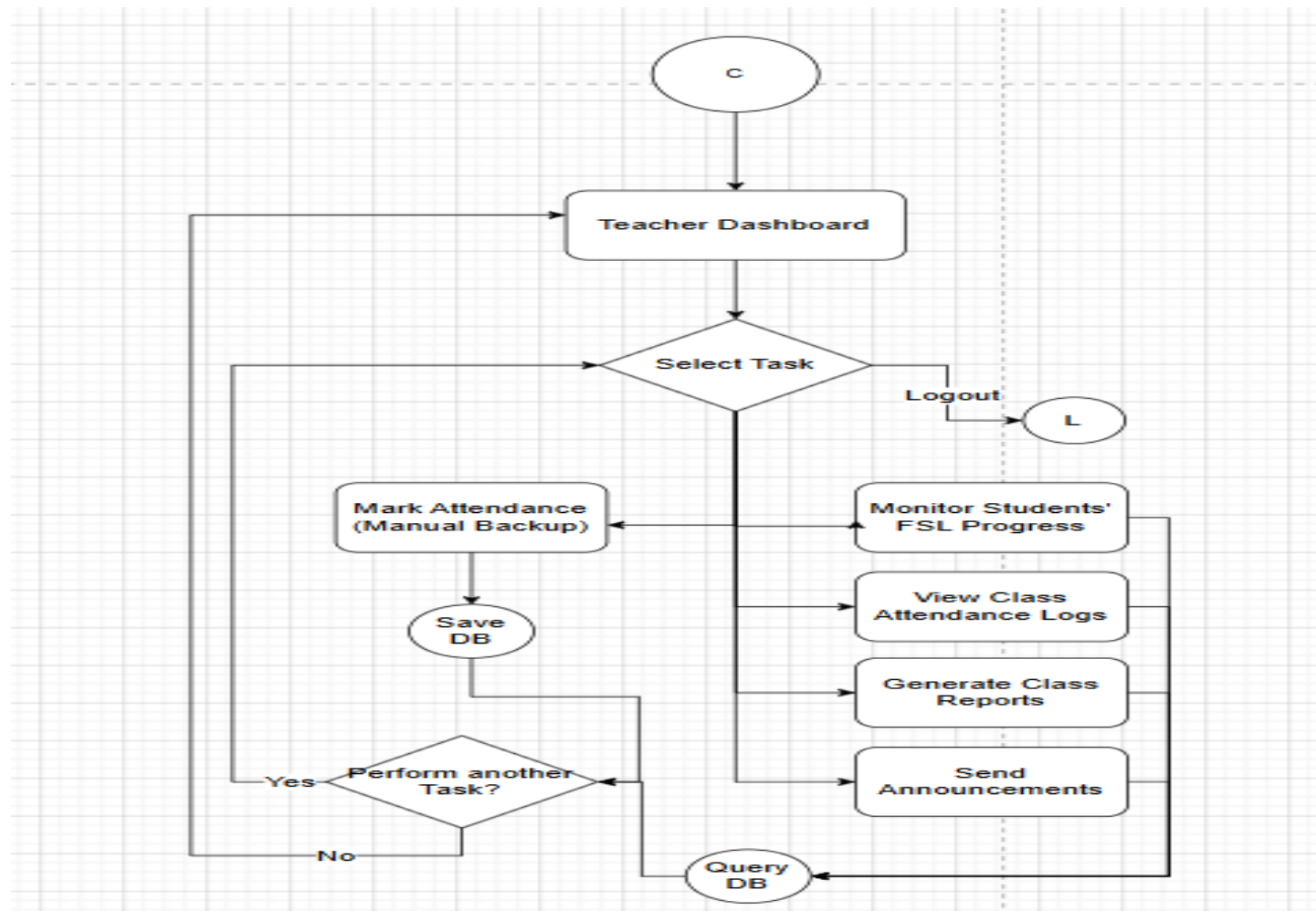


Figure 5 Flowchart (Teacher)

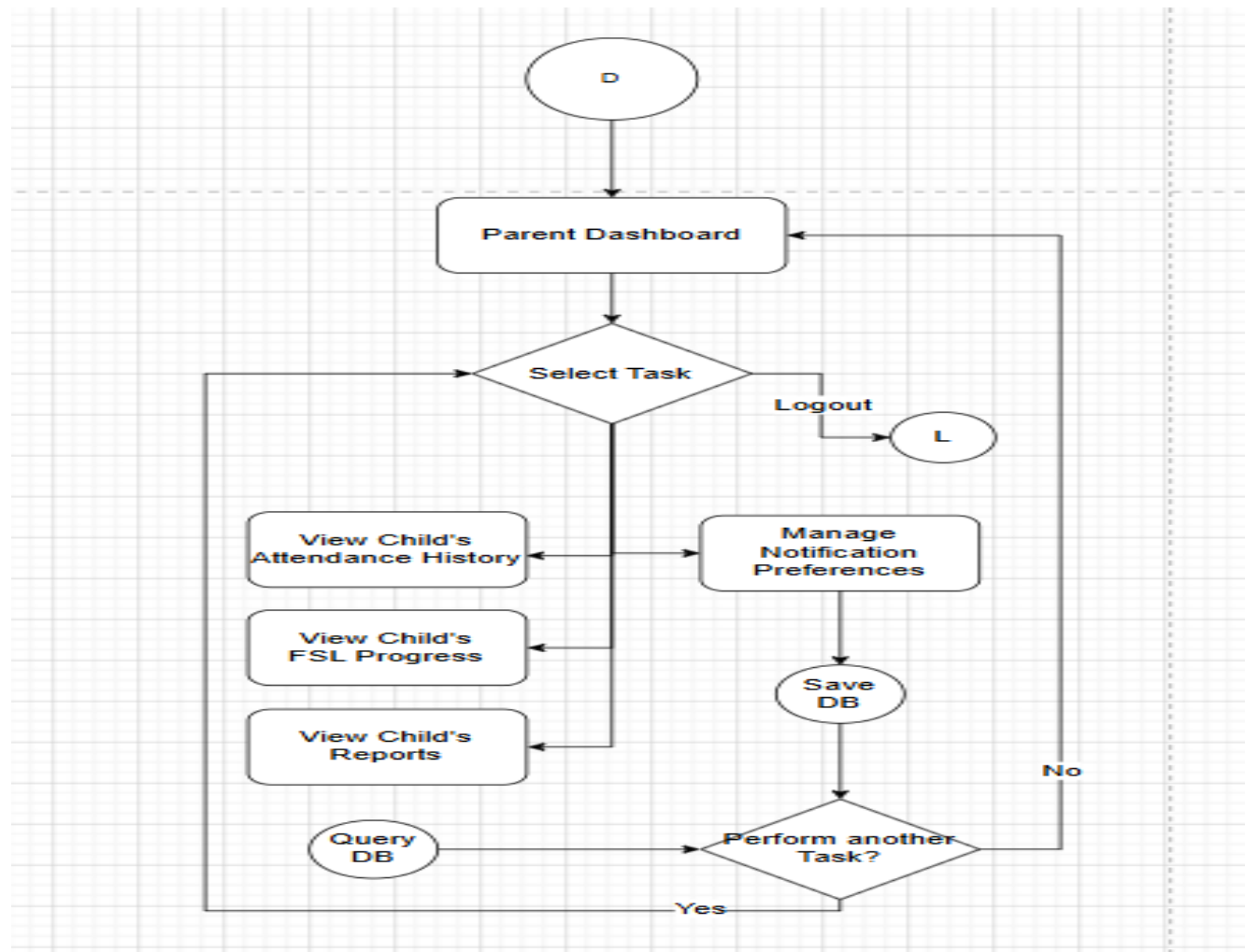


Figure 6 Flowchart (Parent)

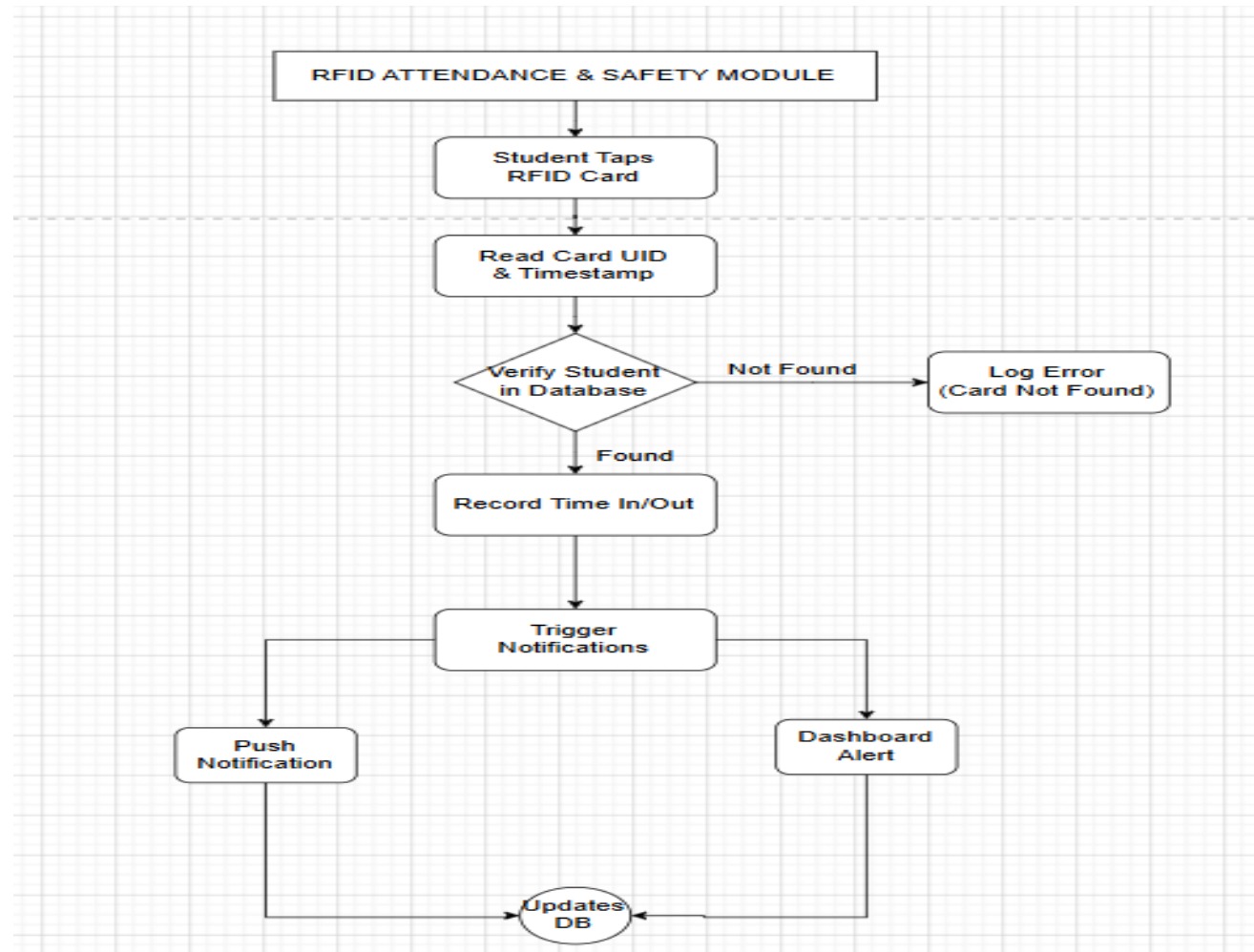


Figure 7 Flowchart (RFID Module)

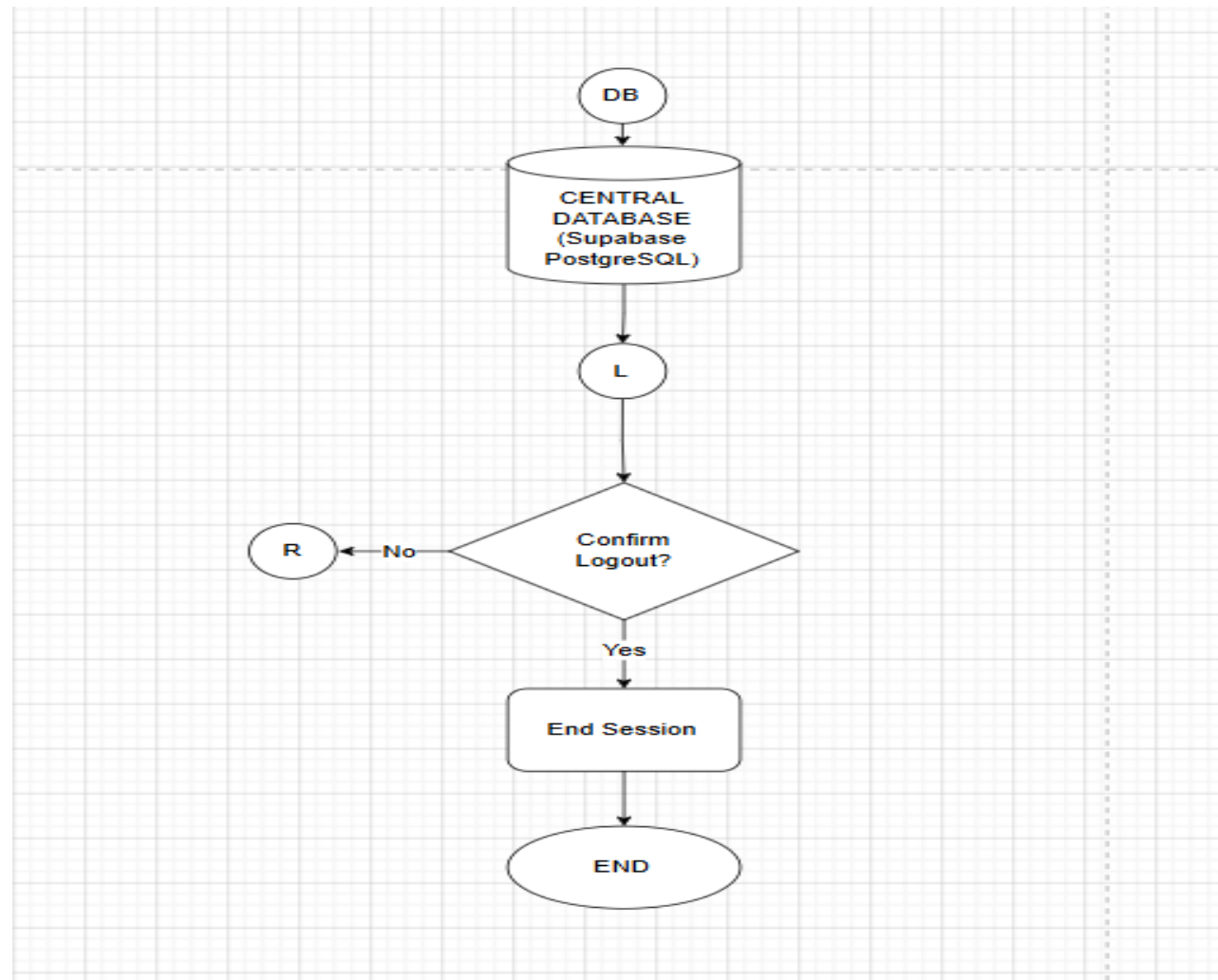


Figure 8 Flowchart (Database)

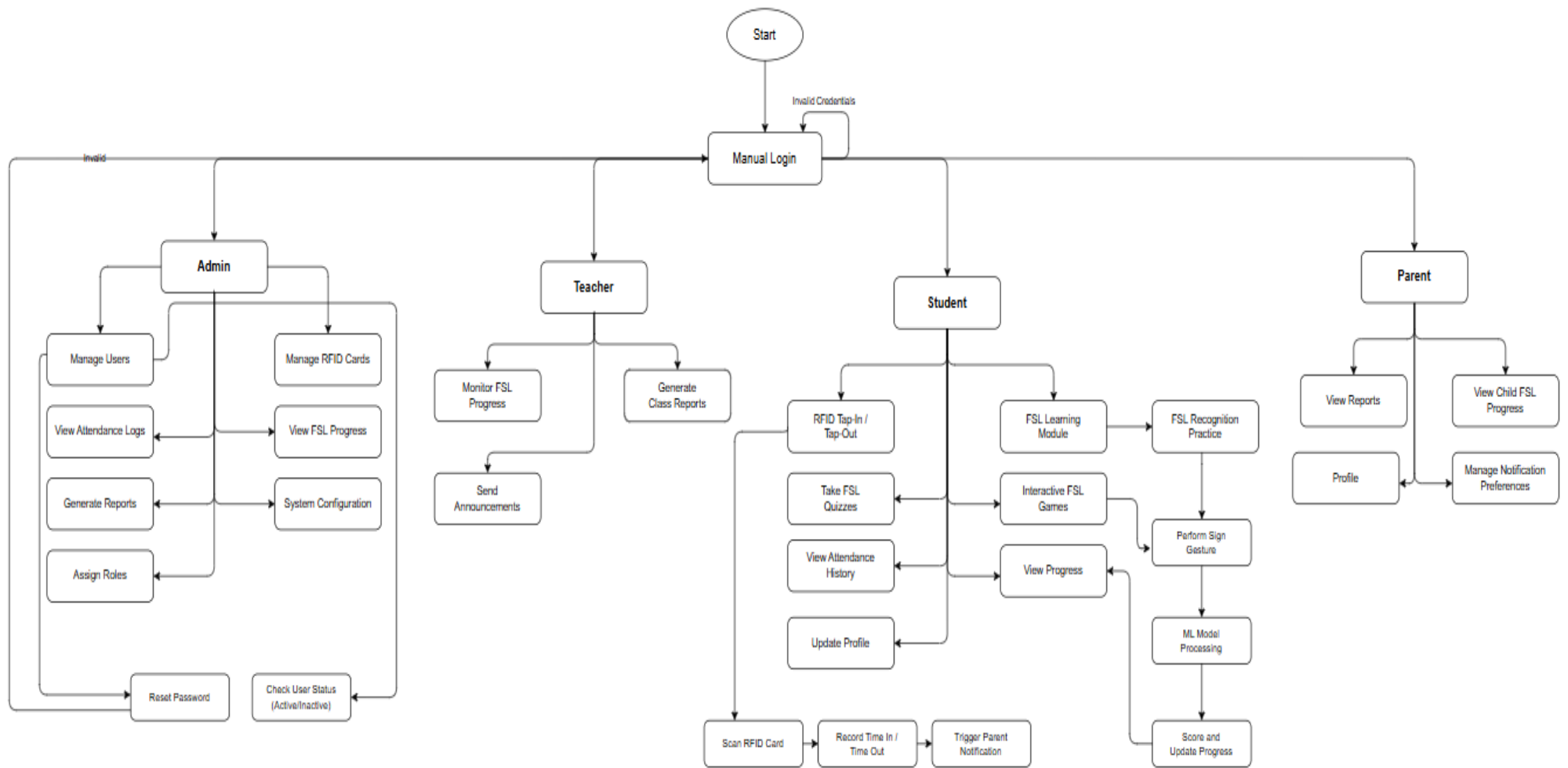
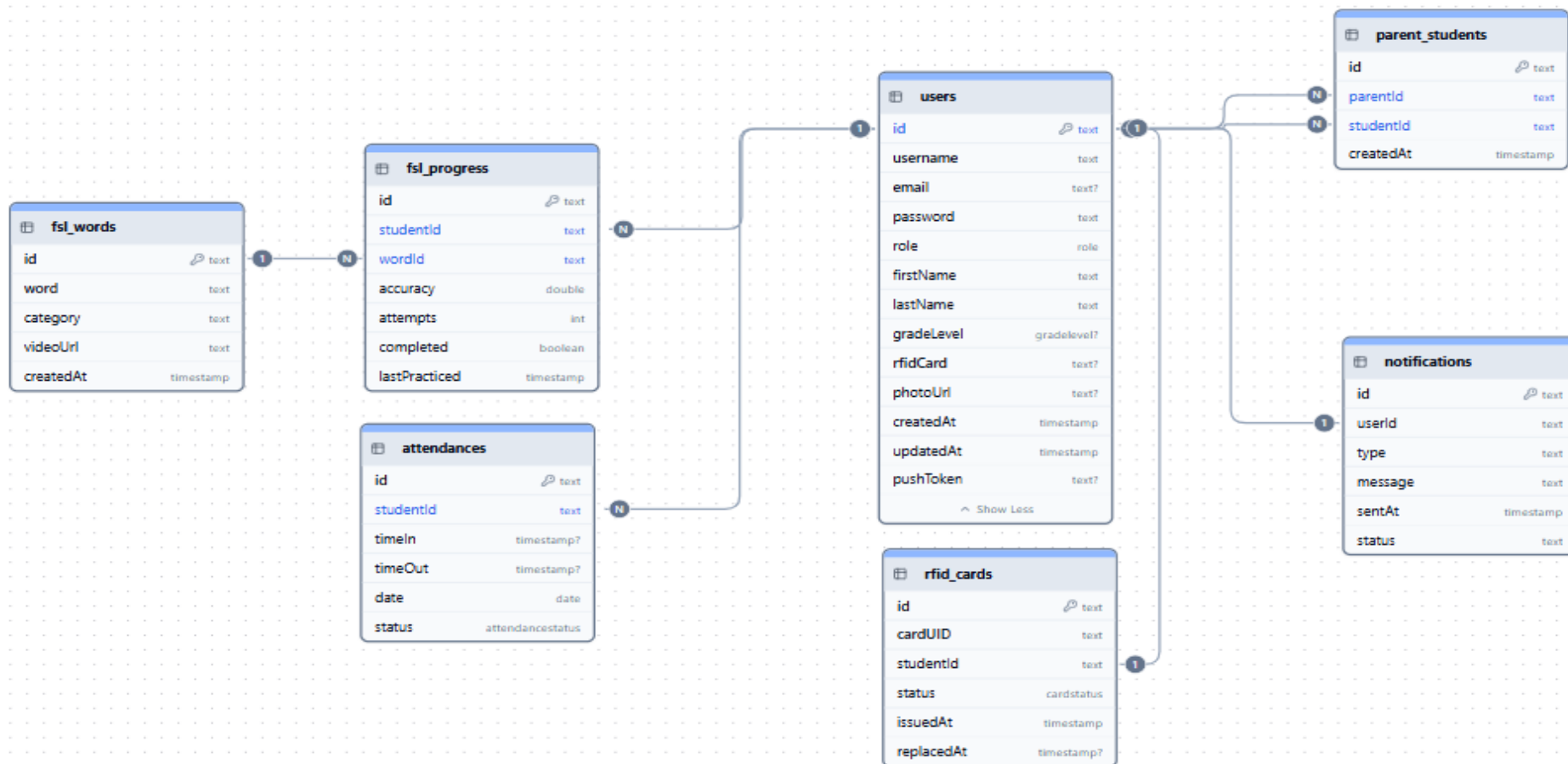


Figure 9 (Data Flow Diagram)

Figure 10 (Entity Relationship Diagram)



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TRANSCRIPT OF INTERVIEW

Here is the full updated interview transcript with all speaker labels properly bolded:

Principal: You have enough time for this project since enrollment and parental consent procedures are still ongoing. Regarding the system, will you also provide the scanner?

Interviewer: Yes, we will provide the system. However, the device we currently have is only a small prototype used for testing.

Principal: How many students are there?

Interviewer: Around 627 students.

Principal: For the RFID cards, will the school provide the cards themselves?

Interviewer: Yes, the cards may be provided by the school.

Principal: What about the scanning device? Will you purchase that as well?

Interviewer: The prototype device we bought is only for testing. We are still determining the recommended scanner for the final setup.

Principal: It would be good if there is a display screen or TV connected to the scanner. When a student scans the card, their photo and name should appear on the screen for verification.

Interviewer: Yes, we will include photo verification so the student's photo and name will appear after scanning.

Principal: Since this will function as an attendance checker, will it also include the student's section, or will it only record a general time-in for the school?

Interviewer: It would be better if the section is included, so the attendance records can be organized by section.

Principal: When do you need the student information? Do you need it by May?

Interviewer: Not necessarily. We are still working on the prototype. Newly enrolled and transferred students usually arrive around June, so the complete list may not be available yet.

Interviewer: We are also planning to create a teacher dashboard where teachers can check and edit attendance records.

Principal: So teachers will have their own dashboard to view attendance?

Interviewer: Yes. Teachers will be able to check student attendance through their dashboard.

Interviewer: We are also considering implementing a QR code attendance system inside classrooms, where students can scan a QR code displayed in the classroom to record attendance.

Principal: We also have a Learning Resource Center that might help provide materials, especially for the Filipino Sign Language component.

Interviewer: That would help us, especially for the FSL recognition system. We need video clips of sign language gestures to train the recognition system.

Principal: You need videos?

Interviewer: Yes. We will be using machine learning, so we need several video samples for each sign.

Principal: How many videos are required?

Interviewer: The recommended amount is 50 videos per sign, but we can work with 10 to 15 videos per sign.

Principal: We have a YouTube channel with validated FSL videos. You can download them from there.

Interviewer: That would be very helpful.

Principal: What student information do you need for the system?

Interviewer: For the RFID attendance system, we will need the student name, LRN, section, parent or guardian name, and parent or guardian contact number. We need the parent contact number because the system will send SMS notifications when the student enters the school.

Principal: So you only need the student's information and the parent's contact number?

Interviewer: Yes, mainly for system integration and account creation.

Principal: Will parental consent be required for this?

Staff Member: Yes, permission can be included during the general assembly or through a consent form in accordance with data privacy regulations.

Interviewer: We are planning to place the RFID scanner at the school gate since that is the main entrance.

Principal: Yes, there is only one main entrance, so placing the scanner at the gate is appropriate.

Interviewer: When a student taps the RFID card, the system will display the student's photo on the screen, allowing the guard to verify the identity.

Principal: Will the system record time-in only, or both time-in and time-out?

Interviewer: We can include both time-in and time-out records.

Principal: That would be useful because some students leave with their parents or service vehicles.

Interviewer: If many students leave at the same time, we may need multiple scanners to avoid long queues.

Principal: Yes, dismissal times usually happen simultaneously. Elementary students leave around 12:00 PM, high school students leave around 1:50 PM, and senior high school students leave afterward.

Interviewer: Scanning itself is fast — around five seconds, especially if the screen displays the student's photo. However, multiple scanners may help reduce waiting time.

Principal: How will the FSL recognition system work?

Interviewer: We will use machine learning. The system will use MediaPipe to detect hand landmarks representing finger and hand positions. After detecting the landmarks, the system will use TensorFlow.js to recognize the sign language gesture.

Principal: So when a student performs a hand sign, the system will recognize the word?

Interviewer: Yes, but it will only recognize 20 to 30 basic words, mainly beginner-level vocabulary.

Principal: What kind of words will you include?

Interviewer: We will prioritize basic words, especially nouns with direct translations in Filipino Sign Language.

Principal: How will you collect the training videos?

Interviewer: Ideally, we will record one instructor performing each sign multiple times, around 15 repetitions from different angles and lighting conditions. Even short three-second clips are acceptable as long as the sign gesture is clearly visible. The more videos we use, the more accurate the recognition system will become.

ICT Coordinator: You are proposing an RFID system for student IDs where students tap the card when entering or leaving the school and parents receive a notification.

Interviewer: Yes. The system will send SMS notifications to parents when their child enters the school.

ICT Coordinator: What information will you need from the school?

Interviewer: We will need the student name, LRN, section, parent or guardian name, parent or guardian contact number, and student photo for verification.

ICT Coordinator: Students already have IDs with barcodes. Will your system use those?

Interviewer: Currently, we plan to use a separate RFID card, similar to a tap card used in trains.

ICT Coordinator: What happens if a student loses the card?

Interviewer: We are considering alternatives, such as RFID stickers attached to the student ID, but we still need to discuss this with our adviser.

Interviewer: The system will include several dashboards: an admin dashboard for system management, a teacher dashboard to view attendance records, a parent dashboard to monitor their child's attendance and progress, and a student interface for the FSL learning tool.

Principal: Do you have a target date for collecting the student data?

Interviewer: Not yet, since we are still in the prototype stage. For now, we only need permission to access the student data once the system is ready.

Principal: If permission is granted, we can send the necessary information through email.

Interviewer: That would be fine.

Interviewer: What gaps or challenges made you decide to propose the FSL project? Is there something lacking at the Philippine School for the Deaf?

Principal: Yes, we lack learning resource materials. That is why we need digital versions of these materials to help improve the literacy of deaf children. The FSL digital version includes expressive signing skills where someone performs sign language, and afterward, students are asked questions about the selection or story presented. Since it is visual, it can help students improve their reading skills.

Interviewer: Thank you, sir.

Principal: You're welcome.

Interviewer: How do the other staff members communicate with the students? How can the project help them?

Principal: The staff members are required to know basic sign language. The digital version of FSL that you are developing can help not only the students but also the staff.

It can help them learn sign language so they can communicate effectively with deaf and hard-of-hearing learners.

Interviewer: Regarding the RFID attendance system, why did you include push notifications? Were there previous issues related to attendance or tracking students?

Principal: The RFID system can help notify parents when their children have already gone home. There have been cases in the school where parents were looking for their children, thinking they were still inside the school when they had already left. With the notification system, parents can be informed about the whereabouts of their children. For security reasons, that is why we want to implement the RFID system.

Interviewer: Thank you very much.

Principal: Thank you as well.

Interviewer: Thank you. So, you mean the system will be used for both research and practical use?

Respondent: Yes, for both. It will involve sign language, specifically Filipino Sign Language (FSL), and will be developed as a web-based and mobile application.

Interviewer: Will it be cross-platform?

Respondent: Yes, it will be cross-platform, with features like push notifications and a mobile application. The goal is to support real-time translation.

Interviewer: Real-time translation of sign language?

Respondent: Yes, through gesture or motion recognition. It can serve as a learning tool — for letters, numbers, and basic words. Machine learning can be used to recognize hand movements and gestures.

Interviewer: So it focuses on gesture recognition?

Respondent: Yes, gesture and movement recognition. Though implementing it may require permissions and resources.

Interviewer: Will there be notifications like SMS?

Respondent: No, not SMS. It will be automatic text-based notifications within the system.

Interviewer: What about limitations?

Respondent: One limitation is language complexity. Sign language is visual and does not directly correspond to spoken or written language. It has its own structure and grammar.

Interviewer: So direct translation may not always be accurate?

Respondent: Yes, it depends on the context. Some words or objects may not have direct translations.

Interviewer: What about basic vocabulary, like family members?

Respondent: Those can be included, like "mother," "father," or "grandmother." But again, FSL does not always match English or Filipino grammar.

Interviewer: How about sentence construction?

Respondent: It's different from spoken language. You may need a system that suggests sentence structures or guides users step by step.

Interviewer: Like a learning interface?

Respondent: Yes, possibly with interactive features like dropdown suggestions, structured lessons, and guided practice.

Interviewer: Will there be progress tracking?

Respondent: Yes, progress tracking can be included, especially for students. It may also include short stories or exercises.

Interviewer: What kind of learning materials?

Respondent: Supplementary materials like vocabulary exercises, short stories, and interactive content. For example, users can follow a story and interpret signs.

Interviewer: Will it be interactive?

Respondent: Yes. If you provide scripts, the system can translate them into sign language visuals. You can design it creatively with images or animations.

Interviewer: Should we use static images or animations?

Respondent: It depends, but animations or videos are better for sign language since it involves movement.

Interviewer: What is your main advice for this project?

Respondent: Make sure the system is useful for the intended beneficiaries, not just for compliance. Focus on practicality and real user needs.

Interviewer: Can we include features like Q&A or story-based learning?

Respondent: Yes, that would be helpful. You can also include picture-based learning, like asking how to sign certain shapes or objects.

Interviewer: Thank you so much.

Respondent: Thank you.

Interviewer: I have a question — regarding school hours, what time do classes start?

Respondent: For the offices, it's 7:00 AM to 4:00 PM, and for classes, it's 8:00 AM to 5:00 PM.

Interviewer: Is the RFID system intended for everyone?

Respondent: Yes, it is for everyone.

Interviewer: How many students are there in total?

Respondent: There are 600 students, from kindergarten to senior high school.

Interviewer: Is there Wi-Fi available here?

Respondent: Yes.

Interviewer: Is it available at the entrance?

Respondent: Yes, both at the entrance and exit.

Respondent (*additional details*): There are 186 elementary students, 236 junior high students, and 205 senior high students.

Interviewee: Actually, I already have something in mind.

Interviewer: What is it?

Interviewee: I would like to create a script, then someone will perform sign language. After that, the viewer will interpret the sign and choose the correct answer from options A, B, C, or D.

Interviewer: Ah, okay.

Interviewee: Something like that.

Interviewer: I also have an idea here, a quiz similar to that, but with basic questions about gestures.

Interviewee: What I have in mind is more focused on receptive skills, because what you are describing is more expressive.

Interviewer: Yes, it is more like visual learning. So maybe we can replace it with that idea instead.

Interviewee: Yes.

Interviewer: Instead of just a generic quiz.

Interviewee: Yes, that would be more applicable.

Interviewee: What I want to happen is a training program not only for children, but also for teachers. For example, in call centers or BPO companies, there are exams that you need to pass before receiving a certificate. You won't be given access if you fail. There is usually an introduction or discussion first, and afterward, there is a quiz. Then participants are asked questions like "What is this?" with objective-type answers: A, B, C, and D. That is what I want to achieve.

Interviewer: So the assessment would consist of sign demonstrations followed by questions, and then they would receive a final grade?

Interviewee: For example, Level 1 (Basic): it will include discussions such as "mother," "father," or basic sentences, followed by an exam. If they pass, they proceed to the next level — Intermediate — where the same process happens: discussion, then exam. After passing that, they move on to Advanced. It should be ladderized.

Interviewer: I see.

Interviewee: Are you capable of implementing that?

Interviewer: The concern is that you mentioned this is for teachers?

Interviewee: Actually, it can also be for children. I only mentioned teachers as an example.

Interviewer: That is possible.

Interviewee: This is not about typing skills — it is about quality. It focuses on cognitive understanding, like "What is being communicated?" or "What is the concept behind it?" and then choosing the correct answer.

Interviewer: Similar to how call centers also have behavioral tests?

Interviewee: Yes, but this one is purely cognitive and focused on Deaf culture.

Interviewer: So should we also create a version specifically for teachers?

Interviewee: No, do not focus on teachers. Just focus on learners (children or general users).

Interviewer: So we will create that system for learners?

Interviewee: Yes. I will prepare the script, if that is okay.

Interviewer: Yes, that's fine.

Interviewee: For example, I will create a script where the signer asks, "Do you understand?" followed by questions 1–10. We will produce the video and include captions. We will provide those to you, and then you will create the corresponding exam.

Interviewer: Okay, we will try.

Interviewee: Once the user passes, they move to the intermediate level. I will also prepare the content for that. Do you think it is feasible?

Interviewer: It is possible, but we still need to check if it fits within our project scope.

Interviewee: At least up to Level 1 would be fine.

Interviewer: It is still aligned with our system since we are also planning to include levels, although the exact flow has not been finalized. If you can provide a sample, we can use that as our starting point.

Interviewee: For example, basic action words (*shows a video*). The "discussion" I am referring to involves users watching a video. Then for the exam, they will simply identify the sign by choosing the correct answer.

Interviewer: Like identification?

Interviewee: Yes, something like that. Is that possible?

Interviewer: Yes.

Interviewee: It is similar to your current module, but instead of static images, it will use videos. We need moving visuals, not just pictures.

Interviewer: Will you provide the script?

Interviewee: Yes. We will conduct a video shoot based on the script, and then you will develop the system. It will become an FSL learning tool.

Interviewer: Okay.

Interviewee: The vocabulary here should not be limited to simple numbers like 1–20. It should be extensive — for example, months from January to December, action words, adjectives, and nouns.

Interviewer: If it is just answering questions, that is manageable. However, if the system needs to recognize signs automatically, that would be more difficult.

Interviewee: There is no need for recognition in the system. We will handle the evaluation manually. The user will simply answer. That is easier than implementing recognition, so just do it that way.

Interviewer: However, we still need to push this module (*referring to their current system component*).

Interviewee: Yes, include that as well, since it only involves letters.

Interviewer: Regarding attendance, what is the current process per classroom? Is attendance recorded manually for students?

Interviewee: Yes, it is manual. For example, the teacher simply checks since there are only a few students — around 10 per section. However, it is not recorded in the official DepEd forms because once it is formally recorded, the student may be marked as dropped. Some students only attend once a week, and it would not be appropriate to fail them immediately if they occasionally show up. To make it easier for your system, what we really want in terms of attendance is simply a record of what time a student leaves or returns to the school.

Interviewer: What about attendance by classroom?

Interviewee: No, do not include attendance by classroom — it would be too complicated. Access to this system should only be given to security personnel, the guidance office, and the principal's office.

Interviewer: So we will remove the "by classroom" feature?

Interviewee: Yes, remove it.

Interviewer: Once the system is deployed, is there an IT team that will handle it?

Interviewee: Yes, we have IT teachers.

Interviewer: For notifications, is SMS really required?

Interviewee: Why? Is it complicated?

Interviewer: Some say it can be costly since there is a charge per message, and we cannot fully test it. That is why we are currently focusing on mobile push notifications instead.

Interviewee: So users will check the app themselves? In that case, let's go with that option since it is more cost-effective.

Interviewer: Regarding the mobile APK download, it has not been finalized yet.

Interviewee: Is it for iOS or Android?

Interviewer: We might have difficulty releasing it on iOS since it requires an Apple Developer account.

Interviewee: Then use Android instead — it will be easier.

Interviewer: Yes, for Android we can just distribute the APK file.

Interviewee: For the FSL letter recognition feature, you should probably remove one part and instead turn it into an FSL tutorial or learning module.

Interviewer: So it will be merged into a single module?

Interviewee: Yes, something like that. We can integrate all our training tools there, focusing more on receptive learning.

INTERVIEW CONFIRMATION

I hereby confirm that the statements and information provided in this interview transcript are true and

accurate based on the discussion conducted.

Interviewee: Jordan Madronio

Signature: 

Date: May 05, 2026